



Impact of example-based XAI for neural networks on trust, understanding, and performance

Maya Perlmutter^{*}, Ryan Gifford, Samantha Krening

The Ohio State University, 1971 Neil Ave, Columbus OH 43210, USA

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ABSTRACT

The purpose of this study is to examine the impact of an example-based explainable artificial intelligence (XAI) interface on trust, understanding, and performance in highly-technical populations. XAI studies often focus on general users in low-risk domains. This study examined the impact of showing the closest matches from the training data from two classes on trust, understanding, and performance for highly-technical users in a high-risk domain. We found that providing example-based explanations significantly increased trust and understanding without decreasing performance. Showing the most similar examples from two classes increased trust more than showing examples from only one class. Participants did not treat different classes the same. The most important features for predicting how well an interface was understood were the helpfulness of the provided examples and the person's trust in the human-machine team. We found priming of highly-technical participants to be particularly important for running XAI studies to mitigate the fear of their jobs being impacted.

1. Introduction

As machine learning (ML) algorithms become more complex, it becomes difficult for people to understand why the ML made a decision, and when or how much to rely on the ML. This can result in less trust in the system, especially in higher risk fields, because users cannot understand what the machine is doing. Explainable AI (XAI) aims to combat this by providing insight into how the ML made its decisions, with a goal of improving understanding, trust, and adoption of the ML system.

One form of XAI is *example-based XAI*, in which examples from the training data are used in an explanation. Example-based XAI has been shown to help users understand how the AI makes its classifications and where it may have limitations (Cai et al., 2019). Users can use the examples to make comparisons to what they are trying to classify, allowing them to infer the best category. However, prior research has not shown how choosing different examples from the dataset, especially whether examples should come from one vs. two classes, impacts trust, explainability, and performance.

Most XAI research also studies a general population of users. For example, a study on example-based explanations had users work with XAI to classify common objects such as onions (Cai et al., 2019). Commercial systems such as Amazon and Netflix use personalized

recommendations from the general population. However, we suspect that highly-technical users have different needs for XAI systems, which impacts trust, explainability, and adoption differently than a general population of users. For example, in the medical field, AI needs to be designed in a way that it has explanations that are verifiable and understandable for practitioners (Müller et al., 2022). Azari et al. found that incorporating physics knowledge of orbiting spacecraft into machine learning helped boost performance and interpretability (Azari et al., 2020); however, it relies on the scientists' domain-specific expertise, so it would not be understood by general users. Consequently, the level of detail and expectations of an AI system may impact trust differently in technical populations, especially in high-risk domains.

The design of XAI in high-risk tasks with highly-technical populations needs to be studied because of the greater consequences when the ML makes mistakes. Errors could risk lives, the environment, create legal and ethical issues (Müller et al., 2022), or result in the significant loss of money.

In order to study XAI in a high-risk task with a highly-technical population, we chose a population of data analysts from an oil and gas pipeline inspection company. The task was to classify sections of pipe as healthy or not. This is a high-risk task since a misclassification could result in: 1) an environmental disaster if a damaged pipe is misclassified

^{*} Corresponding author.

E-mail address: perlmutter.25@osu.edu (M. Perlmutter).

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as healthy, or 2) the loss of millions of dollars for digging up and inspecting a pipe that is misclassified as unhealthy. The goal is to design an XAI system that is highly accurate and facilitates the user's understanding and decision-making process.

The main research question of this paper is to **determine the impact of showing the most similar examples from one vs. two classes has on trust, explainability, and performance in a highly-technical population on a high-risk task**. We also investigated: 1) whether the quality of the explanation impacted trust and understanding, and 2) what factors influence trust in a technical domain that uses a machine learning system.

This paper aims to examine factors that increase a human's trust and understanding in XAI, with a focus on highly technical populations. Neural network algorithms have previously been viewed as less explainable because they include many hidden layers, so people are unable to see which layer or feature the algorithm used to make its decision (Guidotti et al., 2018). This lack of visibility into decision-making often makes it difficult for humans to trust the agents. The algorithm used in this study strives to mitigate these issues by providing examples from the training set as a form of example-based explanation. Specifically, the explainable interfaces will show the user how the algorithm classifies input data based on the closest matches that it finds in the training data. This will provide the user with insight into the features that the algorithm is using to make its classifications. We expect that the increased explainability will increase trust in the machine without harming performance compared to an interface that provides no explanations.

Our main contributions include: 1) Adding example-based XAI that shows examples from multiple classes can improve the human experience without harming performance, which is opposed to the common belief that there is a tradeoff between explainability and performance (Došilović et al., 2018); 2) Highly-technical populations do not treat classes the same, so XAI and interfaces that treat classes the same may be flawed; 3) The main factors that impacted trust in this high-risk domain with highly-technical users were: showing examples from multiple classes, priming, training, unrealistic expectations, interface presentation, and machine errors.

The rest of this paper is organized as follows. First, we discuss the background research, followed by the algorithm design, study design, and procedure. We then report our results for how example-based explanations impacted trust and performance, and we describe qualitative results from free response questions.

2. Background

2.1. Explainable artificial intelligence

Deep learning algorithms have become increasingly popular in recent years due to successes with algorithms such as AlphaGo and their accuracy with high-dimensional data (DeepMind 2015). However, they are often black box algorithms that provide no explanation as to how they make predictions, so users have no way of knowing whether or why the AI made an error. For example, Caruana et al. found that a neural network predicted that patients with asthma had a lower risk for pneumonia (Caruana et al., 2015). It was only after using a rule-based algorithm showing explanations that they learned the algorithm classified asthma patients as lower risk due to the aggressive treatments and shorter stay in the hospital. Without the rule-based explanation, the researchers would have had no insight into the algorithm's mistaken decisions, and patients with asthma would have been misclassified and put at greater risk. Another problematic instance is a Tesla car that identified the moon as an orange spotlight and started braking on the highway (Levin, 2021). Drivers need to understand when and how the sensors will make errors for safety.

Many researchers have worked to mitigate these black boxes by introducing explainable artificial intelligence (XAI) algorithms to

increase explainability. XAI enables users to understand something about the machine learning agent or how it made its decisions. Hoffman et al. expand on these metrics by noting that the XAI must be able to help people develop good mental models of how the system works along with when, how, and why it can fail (Hoffman et al., 2022). One issue with XAI is that it is a rapidly evolving field where there are different definitions being used across disciplines (Schneeberger et al., 2023). In this work, we define XAI as any AI that provides some explanation as to how or why it made its decision (Adadi and Berrada, 2018).

A goal of the XAI field is to gauge how understandable or interpretable an XAI system is. Gilpin et al. defined interpretability as the ability to describe the inner workings of a system in a way that can be understood by humans (Gilpin et al., 2018). Similarly, Doshi-Velez and Kim define interpretability as the ability to explain or present in understandable terms to a human (Doshi-Velez and Kim, 2017). These goals align with the idea that simply having an explanation is not enough. The human must be able to understand those explanations; otherwise, the system is not explainable, and it will not be helpful for the human. Understanding could be linked to common ground, since humans and machines need to understand the shared knowledge, beliefs, and goals shared among them to succeed when working on tasks together (Klein et al., 2005). Enhancing understanding in the machine can result in higher trust (Lee and See, 2004; Chiou et al., 2022). Thus, it may also be important to consider understanding along with trust when investigating explainability.

There are many strategies for explainability (Adadi and Berrada, 2018). Globally interpretable models provide insight into the entire model, such as Caruana et al.'s algorithm that predicted pneumonia risk (Caruana et al., 2015); whereas locally interpretable models, such as Ribeiro et al.'s LIME, provide explanations for individual predictions (Ribeiro et al., 2016). Model-specific methods only apply to specific algorithms such as decision trees (Krishnan et al., 1999) or linear regression, while agnostic methods can be applied to any algorithm, such as rule-based explanations and example-based explanations (Caruana et al., 2015; Kim et al., 2014; Bućinca et al., 2020). Intrinsically explainable algorithms, such as decision trees and linear regression, provide visual explanations to enable users to quickly understand the ML predictions. However, these algorithms are of limited use because they cannot handle complex data, so they do not scale well to real-world problems (Krishnan et al., 1999; Sarkar et al., 2016). Explanations can take many formats, including visual representations such as heatmaps, graphs (Yang et al., 2020), examples (Cai et al., 2019), images (Bućinca et al., 2020), or they can have verbal forms such as decision-trees or rules (Caruana et al., 2015; Krishnan et al., 1999).

Researchers have examined several ways to evaluate XAI. Zhou et al. published a survey on metrics and found that metrics such as trust and confidence were the focus of human-centered evaluations, and that quantitative metrics such as clarity, broadness, soundness, parsimony, and completeness are used to evaluate the interpretability and fidelity of explanations (Zhou et al., 2021). Holzinger and Müller introduce the concept of causability, or the measure of whether and how much users can understand an explanation (Holzinger and Müller, 2021). Müller et al. take the idea of causability a step farther and apply it to developing XAI for In Vitro Diagnostic devices in the context complying to medical AI regulations (Müller et al., 2022). While we do not directly use these evaluations, we use primarily human-centered metrics such as trust and understanding to evaluate our XAI and we include performance metrics to determine the effectiveness of the explanations. The rest of our work will focus on example-based explanations.

2.1.1. Example-Based explanations

Example-based explanations, sometimes called prototype XAI, present examples that are taken from the training dataset to explain the behavior of the model (Kim et al., 2014).

It is an open area of research to determine which examples should be chosen from the dataset. The Bayesian Case Model (BCM) performed

unsupervised clustering and generated the most representative examples and features, finding that it maintained accuracy while increasing interpretability (Kim et al., 2014). Instead of choosing the most similar examples, MMD-critic also chose examples that fail to represent the data well (Kim et al., 2016). Kim et al. also found that participants performed best in a classification task when they were presented with examples along with criticisms rather than examples alone (Kim et al., 2016). A gap in the literature we address is whether examples should be shown from two classes for classification tasks.

Cai et al. found that participants who were shown example-based explanations felt they had a better understanding of the system when asked to classify household objects such as onions (Cai et al., 2019). Although these examples provide greater insight into the algorithm, they can also reveal limitations by presenting examples from categories that may be unrelated to a target image in a classification task (Cai et al., 2019). While Kim et al. suggest that criticisms can help users with classifications (Kim et al., 2016), Cai et al. claim that the criticisms only reveal limitations of the algorithm (Cai et al., 2019). Explainability can provide users with greater awareness of the algorithm's behaviors and potentially help them understand why the algorithm makes errors, but it is essential to evaluate explainable algorithms to ensure that they are applied in the most suitable contexts. The XAI may need to be tailored to the application and user population.

Even so, these research teams only examined whether example-based explanations were viable for black box algorithms; they did not examine these explanations in the context of trust or understanding. Bućinca et al. directly compared different types of XAI, and found that trust varied with task completion, suggesting that context impacted which XAI was more beneficial (Bućinca et al., 2020). None of these studies varied the number of classes in the explanations or recruited participants in technical populations and high-stakes domains. Hegde et al. developed Similar Medical Images Like Yours (SMILY) a reverse image search tool for histopathology using images as a query, and found that SMILY was able to find images with similar histologic features and organ sites (Hegde et al., 2019). Here, Hegde et al. focus on a higher-stake domain with more technical practitioners; however, they did not evaluate user experience metrics such as trust and they only recruited three participants, so it is unclear whether SMILY's performance would generalize across more participants.

Our study addresses a gap in these findings by examining the impact of one vs. two classes of explanation on trust and performance in a technical population.

2.1.2. Users

The needs of the users are important considerations when evaluating AI. Hoffman et al. provided a series of metrics to evaluate XAI, such as the goodness of the explanations, whether the user understands the AI systems, and how well the human-XAI team performs (Hoffman et al., 2018). Similarly, Mohseni et al. identified three key groups of users—novices, data experts, and AI experts—and highlighted how each group had different needs and levels of understanding that the XAI designers must account for to promote successful interactions with the algorithms (Mohseni et al., 2021). While many of the papers on XAI have focused on novices, our work will examine data experts that regularly interact with AI to determine whether example-based explanations are suitable for technical populations to increase trust, understanding, and performance.

2.2. Trust

In XAI literature, trust has been defined in multiple ways. For example, Lee and See define trust as “the attitude that an agent will help achieve an individual's goals in a situation characterized by uncertainty and vulnerability,” where the agent could be a human or a machine (Lee and See, 2004). Vereshak et al. expanded this definition, noting that trust involves three components: 1) trust requires vulnerability; 2)

positive expectations precede trust; and 3) trust is an attitude rather than a behavior (Vereshak et al., 2021). “Trust” can have a different definition from “trustworthy”. In ISO TR 24,028, “trust” is defined as the degree that a user or stakeholder is confident that a system will behave as intended, while “trustworthiness” is defined as the ability to meet a stakeholders' expectations in a verifiable way (ISO/IEC TR 24028 2020). This definition of trust also relates to Johnson et al.'s concept of predictability, where systems must be predictable enough that they can be reasonably relied on (Johnson and Bradshaw, 2021).

In the context of this work, we follow Lee and See's and Vereshak et al.'s definitions for trust because our study has participants working with agents on tasks where there is uncertainty and they need to be able to rely on the agents. On the other hand, in this study, trustworthiness is ensuring that the design of the XAI meets the analysts' expectations.

2.2.1. Factors that impact trust

Explainability can boost trust because explanations can make an algorithm appear more predictable to a user (Johnson and Bradshaw, 2021). Wang et al. found that higher fidelity and more robust explanations boosted trust (Wang et al., 2023). Angersschmid et al. also found that having AI explanations increased user trust but the type of explanation did not impact trust (Angersschmid et al., 2022). Unlike Angersschmid et al., Yang et al. found that some visual explanations were more beneficial than others (Yang et al., 2020). Specifically, Yang et al. conducted a study on trust and XAI, where they had participants classify leaves with algorithms that provided different visual example-based explanations and found that all the explanation types increased trust but that some visual layouts such as grids were more beneficial (Yang et al., 2020). However, Wang et al., Angersschmid et al., and Yang et al. did not explore the impact of varying whether the explanations were from one or two classes (Wang et al., 2023; Angersschmid et al., 2022; Yang et al., 2020). Dzindolet et al. found that people were more likely to trust an agent, even after it made mistakes, if they were able to understand how the agent made its decision (Dzindolet et al., 2003). However, they only provided a generic explanation at the beginning of the task, and the explanations were verbal rather than visual, so the results may not generalize to XAI that provides explanations throughout the interactions. One way we address these limitations is by studying a highly-technical population in a high-risk domain, while providing examples from two classes. Ashoori et al. found that participants preferred transparency and human involvement (Ashoori and Weisz, 2019). Zhou et al. found that presenting the influence of training data points on predictions boosted trust, but only for data points with high influence values (Zhou et al., 2019). Arshad et al. found that user confidence is important for trust and that uncertainty can impact confidence (Arshad et al., 2015). These findings all suggest that transparency and having more information or certainty can impact trust and decision-making. Our study shows example-based explanations from one or two classes so that users can have more information before making a decision.

On the other hand, people are less likely to take advice from an agent after it has made a mistake, especially after the agent makes an obvious error (Goodyear et al., 2016; Madhavan et al., 2006). Furthermore, Angersschmid et al. found that lower perceptions of fairness decreased trust, indicating that perceived fairness should be considered when designing for trust (Angersschmid et al., 2022). This decrease in trust may then adversely affect the performance of the team.

2.3. Human-Machine team performance

We define a human-machine team as a human working toward a common goal with an AI or other agents (O'Neill et al., 2022). While the accuracy of the algorithm is seen as one of the most important features of ML, performance is more than the theoretical accuracy of the ML. People achieve a higher performance when working with an ML agent they have a positive experience with (Krening and Feigh, 2018). Design features like whether the human can provide input about the future and

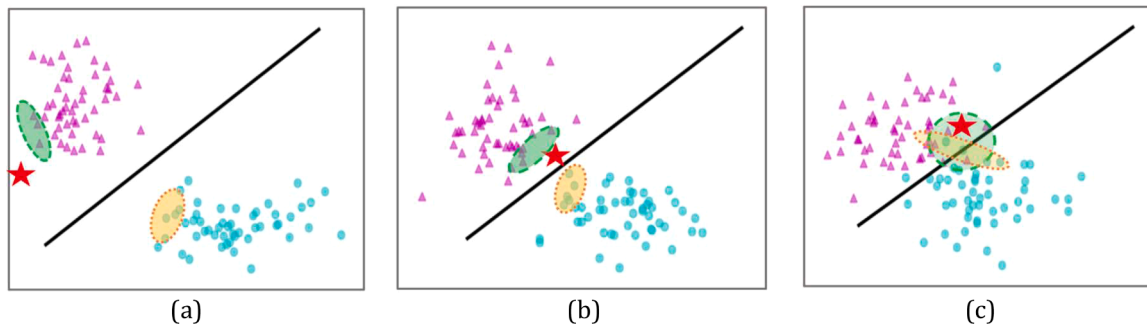


Fig. 1. Choosing examples for XAI.

how quickly the ML provides feedback to the human impact the objective performance of the human-machine system (Krening and Feigh, 2019). The AI should facilitate decision-making while also helping users understand when and when not to use the AI so that they do not overrely on it (Rayo et al., 2020). The type of explanation can impact performance (Bućinca et al., 2020). Even so, there is no human-machine team if the user does not choose to adopt the system. Thus, we examine the impact of XAI on trust and performance.

3. Methods

3.1. Choosing examples from one vs. two classes

Example-based XAI research has not thoroughly determined which examples should be shown to the user along with the impact the chosen examples can have on the user's trust in the ML system. To explore this, we created an XAI system that could show participants the most similar examples from the dataset from two classes.

Consider the toy example in Fig. 1. The scatter plots show data in two dimensions. There are two classes: magenta triangles and cyan circles. Assume that the red star was not in the training data and needs to be classified. The ML classifies the red star as belonging to the blue circle class. An XAI system that shows users the closest training data from the magenta class would show all magenta data within the green dashed ovals. An XAI system that shows the users the closest training data from the cyan class would show all cyan data within the orange dotted ovals.

In Case (a) of Fig. 1, the two classes are linearly separable. The red star is relatively far away from the boundary line between the classes. The closest training data from the magenta triangle class, which are inside the green oval, will appear similar to the red star data. The closest training data from the cyan class, which is inside the orange oval, will appear quite different than both the red star and green oval examples. A user looking at the examples in the green and orange ovals will see a large difference between examples. We expect this will help a user understand that the ML sees a difference between classes, which will increase trust in the system.

In Case (b), the two classes are still linearly separable, but the red star is close to the boundary between classes. Even so, there is likely to be a distinct difference between the examples shown in the green and orange ovals. We expect this ability to compare and contrast how the ML views the classes will build understanding and trust.

In Case (c), the classes are not linearly separable, which is more realistic because complex data can be difficult to classify. The closest examples from both classes are very similar – in fact, the ovals are overlapping. When users see the similarity in the examples, they are likely to realize this is a confusing case for the ML. They may then pay more attention and provide more input to the ML to aid the classification process. We expect that showing the user that a specific case is not clear cut will increase trust because they can understand the limits of the ML. It is more ideal than an ML system that reports a classification 'certainty' estimate that turns out to be wrong. We hypothesize that showing users



Fig. 2. Baseline interface.

the closest training data from two classes is a more nuanced method of communicating uncertainty of the ML's classification to the user.

3.2. Interface and ML design

To predict whether an image of a pipeline section was considered normal or an anomaly, a Convolutional Neural Network (CNN) was used. The CNN was trained for 100 epochs using cross-entropy loss and the ADAM optimizer. Images of 100×100 pixels were passed through two convolutions, each followed by a batch normalization layer and a max pooling operation that down-sampled the feature space by a factor of 2. The image features were then flattened and fed into 4 fully-connected layers with hidden node sizes of 50, 25, 6 and 2. The last layer used a Softmax activation to output the probability of an image being normal and an anomaly.

To find the closest examples from the training dataset after the convolutional portion of our trained CNN, the activation graph of the four fully-connected layers was treated as the 'fingerprint' for each patch. A PNG of the neural network connectivity graph was created, where the activation level of a node determined the alpha level of its edge that connected it to the next node in the graph. Only non-zero activations were visible in the graph, as zero values meant no signal from that node was fed forward into the next layer of the graph (ReLU activation ensures only positive values). Once created, the key points were found for each activation graph using the Harris Corner Detector algorithm. The most similar examples were then found by finding the training examples with the smallest absolute difference between the key points of the test image and key points of the training samples.

We describe the interfaces used in this study below and show the interfaces in Figs. 2-4. The color scale matches the existing visualization tool used by the analysts.

Interface Baseline: The participant only saw the current input image and the suggested classification by the model. The interface did not provide any further explanation. See Fig. 2.

Interface XAI-1: The XAI showed closest matches within one class.

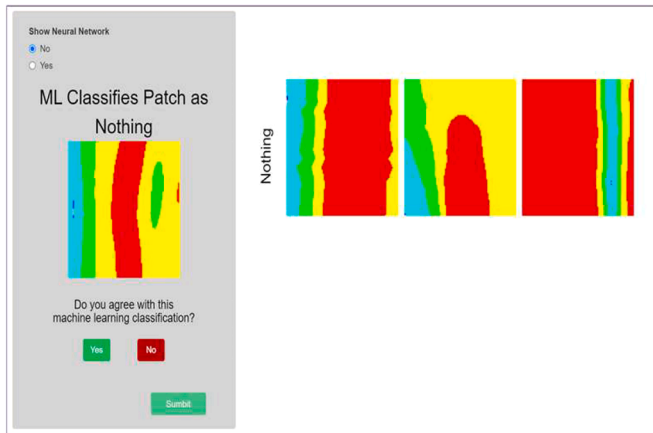


Fig. 3. XAI showing closest matches within one category. (Interface XAI-1).

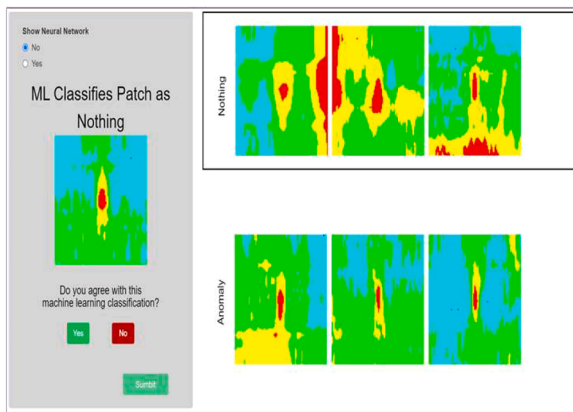


Fig. 4. XAI showing closest matches in both categories. (Interface XAI-2).

For each case, the ML classified an image. The participants saw the current input image, the suggested classification by the algorithm, along with the closest matching images from the suggested class in the training data. For example, if the algorithm predicts that an input image is an anomaly, then the explanation would show three of the closest cases of anomalies from the training data. This is like showing the yellow oval examples to a user from Fig. 1. See Fig. 3 for the interface layout.

Interface XAI-2: The XAI showed closest matches in both classes. The participants saw the current input image, the suggested classification by the algorithm, along with three examples of the closest matching images from the training data for both classes. This is like showing both the yellow and purple oval examples to a user from Fig. 1. See Fig. 4 for the interface layout.

3.3. Participants and design

We conducted a within-subjects study in which we investigated the effect of three interfaces on the human's experience with the agent. Our 24 participants were data analysts from a pipeline inspection company

Table 2

A list of questions and scales used in the human experience questionnaire (Step 2b).

Question:	Scale (Measured Continuously):
How much do you agree with the ML agent's classification?	0 (Strongly disagree) to 10 (Strongly agree)
How much do you trust the machine learning classification?	0 (Strongly distrust) to 10 (Strongly trust)
How much do you trust the human-machine team (which includes you)?	0 (Strongly distrust) to 10 (Strongly trust)
To what extent did showing similar cases help you determine the input's classification, regardless of whether you agreed with the ML classification?	0 (No help at all) to 10 (Helped a lot)
To what extent do you feel you understand why the ML made the classification it did?	0 (Do not understand at all) to 10 (Understand)

with varying levels of experience at the company. We divided experience level into four groups based on the amount of time they worked as a data analyst at the company: 2 participants in the 6 months - 2 years group; 13 participants in the 2 - 5 years group; 6 participants in the 5 - 10 years group; and 3 participants in the 10+ years group. Since we studied analysts from industry, we had a limited population; a sample size of 24 was enough for a $\beta=0.10$ and a two-tailed $\alpha=0.05$.

Participants were asked to classify oil pipeline images as normal ("nothing") or abnormal ("anomaly"). They were provided with predictions made by a neural network model and were asked whether they agreed with the predictions. Participants interacted with three interfaces that had varying levels of explanation to classify the pipeline images. One interface was a Baseline that provided no explanation, another interface showed the three closest examples from the class that it selected as its prediction, and the final interface provided users with the three closest classifications from both the 'nothing' and 'anomaly' classes.

3.4. Procedure

The procedure followed by all participants is provided below. All participants experienced the interface variations in a randomized order. Table 1 summarizes the procedure.

3.5. Human experience measures

After interacting with each interface, participants completed a questionnaire (Step 2b) about: 1) the extent to which they agreed with the classifications; 2) the extent to which they trusted that algorithm; 3) the extent to which they trusted the human-machine team; 4) whether viewing similar cases helped them understand the agent's classification; and 5) whether they understood why the agent made the classification that it did. These Human Experience measures were measured continuously using a Likert scale, much like the scales used by Wang et al. and Zhou et al. but on a scale of 10 rather than 5 and 9, respectively (Wang et al., 2023; Zhou et al., 2019). Values of 10 indicated strong agreement, strong trust, helped a lot, and strong understanding. Values of 0 indicated strong disagreement, strong distrust, no help at all, and no understanding. The questions from the human experience questionnaire

Table 1

A summary of the procedure.

Procedure with Human-Subjects
1. Informed consent.
2. For each of the 3 interfaces:
a. For each of 3 cases (e.g. trials):
i. Participants were shown the local image of the pipeline and the ML agent's classification. They were asked whether they agreed with the classification.
b. Questionnaire about their experience with the interface.
3. Questionnaire comparing their experiences with the interfaces.

Table 3

A list of objective measures (Step 2ai).

Question:	Scale (Nominal):
Objective Performance	1- The human agreed with the machine and the machine was correct. 2- The human agreed with the machine, but the machine was incorrect. 3- The human disagreed with the machine, but the machine was correct. 4- The human disagreed with the machine, and the machine was incorrect.
Objective Agreement	1- The human disagreed with the machine 2- The human agreed with the machine

are summarized below in Table 2.

After interacting with each interface and at the end of the study (Steps 2b and 3), participants ranked the interfaces by trust and understanding. Participants were also given the option to explain factors that impacted their experience working with the interfaces. For example, participants were prompted, “Please provide comments on what impacted your trust in this interface.” Participants were not primed or given any information for these written responses.

3.6. Objective measures

While participants interacted with the interfaces (Step 2ai), objective performance metrics were logged to data files. Whether the participant agreed with or overrode the ML’s classification, along with whether the resulting decision was accurate, were recorded.

We analyzed our objective measures using ANOVAs and other non-parametric tests such as the Kruskal-Wallis test and the Chi-Square test when the data either had ordinal measures, imbalanced groups, or violated assumptions for ANOVAs. A limited analysis of the narrow research question on trust can be found in our associated conference paper (Perlmutter and Krening, 2023). Table 3 summarizes the objective metrics.

3.7. Machine learning analyses

In addition to our inferential analyses, we used machine learning to investigate complex interactions between features. Specifically, we used Recursive Feature Elimination (RFE), which iteratively removes the

least important feature for predicting a target variable until there is only one variable left. The remaining variable is the most predictive feature. This allows us to rank feature importance while taking complex interactions of features into account. However, it is important to note that in machine learning, prediction does not mean causality; rather it’s the ability of the machine learning to calculate a label.

4. Results and discussion

The results will begin with the Human Experience data from Step 2b and 3, followed by the Objective data collected from Step 2ai. At the end, we discuss free response answers from Steps 2b and 3.

Fig. 5 shows the distributions of the Human Experience measures (Step 2b). A line is drawn at the middling value of 5 for each of the factors. All of the Baseline measures and many of the XAI-1 measures resulted in bi-modal distributions, whereas XAI-2 were mostly unimodal. This indicates that the Baseline and XAI-1 interfaces caused groups of participants to react quite differently, with one set of participants having a worse user experience. However, the XAI-2 interface had a more uniform user experience. Overall, participants understood the XAI-2 interface the most, indicating that it aids understanding to show examples from two classes. Trust in the human-machine team was much higher across all interfaces than trust in the machine alone. Most users found interfaces that showed similar examples to be helpful in understanding the ML, while a few participants found the examples to provide very little help. The perceived agreement of the Baseline and XAI-1 interfaces were bi-modal and almost the reverse of each other; whereas the XAI-2 interface was unimodal and more positive on average. See Fig. 5 below.

4.1. Trust

4.1.1. Trust rankings

At the end of the study in Step 3, participants were prompted: Please rank each interface you trusted in order of most to least. We conducted a Kruskal Wallis test and found a significant difference in ranking between the interfaces ($\chi^2(2) = 11.39$, $p = 0.003$, $\eta^2 = 0.18$). Participants consistently ranked the XAI-2 interface as the most trusted, followed by XAI-1, followed by the Baseline. See Fig. 6.

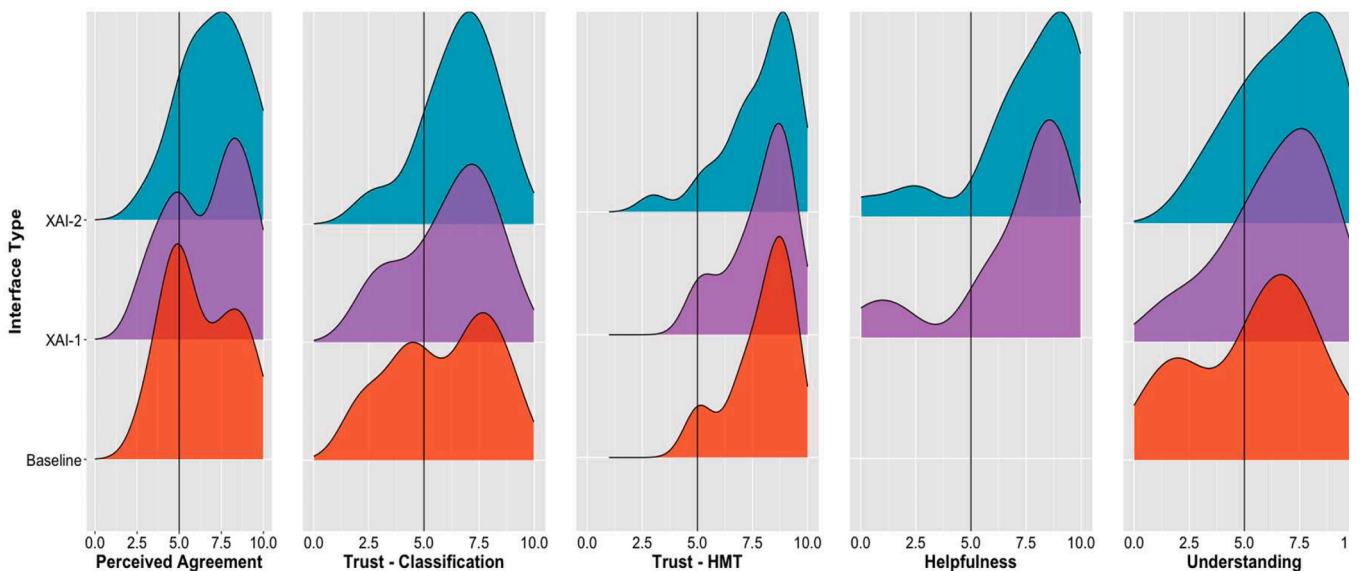


Fig. 5. Distribution of Human Experience Measures (Step 2b). Note: Participants were not asked about helpfulness of the similar examples when interacting with the Baseline interface since similar examples were only shown in the XAI interfaces.

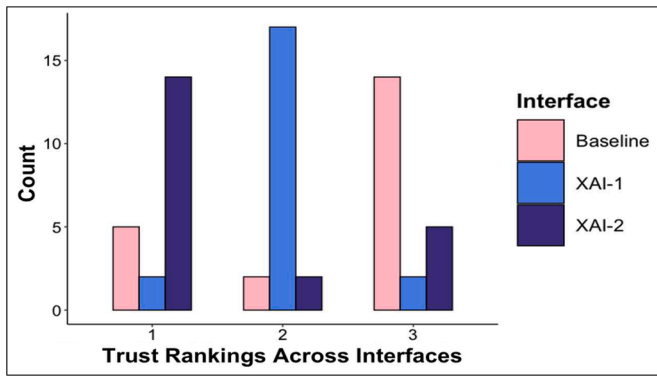


Fig. 6. A grouped bar plot comparing rankings across the interfaces. A rank of 1 indicates that an interface was considered the most trusted. (Step 3).

4.1.2. Trust in the classification

Immediately after interacting with an interface (Step 2b), participants were asked: How much do you trust the machine learning classification? We wanted to study whether participants responded differently regarding trust immediately after interacting with an agent compared to after they interacted with all agents. After splitting the data by interface, we found that assumptions for normality and homogeneity were met so we were able to use an ANOVA. We did not find significant differences in trust in the ML's classifications by interface from the Step 2b data, even though they ranked the XAI-2 interface significantly higher in trust at the end of the study ($F(2, 69) = 0.719, p = 0.491, \eta^2(2) = 0.020$) (Step 3). It is possible that the effect size was smaller for the Step 2b data since it did not directly compare interfaces. Another possibility is that participants viewed the ML and XAI interfaces as different entities, which garnered different levels of trust.

Also, a Kruskal-Wallis test found that there were no significant differences in trust in the ML's classification by years of experience (Step 2b, $\chi^2(3) = 3.43, p = 0.330, \eta^2 = 0.048$). All participants had a moderate- to high-level of trust in the machine regardless of their level of experience. This finding differs from that of Arshad et al.'s who found that there were differences in user experience based on expertise (Arshad et al., 2015). However, our findings focus on trust rather than confidence and our groups were imbalanced, so future work will address this.

Our next step was to determine which features impacted trust in the

classification, especially since interface was not significant in the Step 2b data. For each interface, we trained a Random Forest Regression model to predict trust in the ML's classifications using years of experience, objective performance from Step 2ai, Step 2b data (including perceived agreement and understanding), and Step 3 data. The r^2 value for each model was: Baseline $r^2 = 0.95$; XAI-1 $r^2 = 0.95$; XAI-2 $r^2 = 0.90$. The **most important feature for predicting trust** in the ML's classifications for all interfaces was the participant's **perceived agreement** with the classification. For the XAI-2 model, the participant's nervousness against robotics was also an important factor in trust, although much less than perceived agreement (i.e. *I feel that if I depend on robots of artificial intelligence too much, something bad might happen.*)

4.1.3. Trust in the human-machine team

In Step 2b, participants were asked: How much do you trust the human-machine team (which includes you)? Interestingly, there was not a significant difference between interfaces when examining trust in the human-machine team. However, we found that people had higher trust in the human-machine team than in the machine learning agent alone. We found that both trust in the machine and trust in the human-machine team violated assumptions of normality, so we conducted a Spearman's Rho correlation. There was a significant and positive Spearman's Rho correlation between trust in the machine and trust in the human-machine team across all interfaces ($r(70) = 0.527, p < 0.001$). Additionally, we conducted a Wilcoxon Signed Ranks test and found that trust was rated significantly higher for the human-machine team compared to the machine's classifications ($Z = -6.186, p < 0.001$). A Cohen's effect size of $d = 0.73$ suggests a moderate amount of significance. These findings indicate that **the ability for a human or override or contribute to the decision is a more important factor for trust than the examples shown** in an interface's explanation. See Fig. 7.

Users in technical domains may have a greater need for control over the ML because they work on high stakes problems. In this case, the personnel at the oil and gas pipeline inspection company could risk environmental disaster with pipe leaks, so they may want to be more involved in the decision-making process rather than rely on an ML model. Beyond oil and gas pipeline inspection, the same need for control may be more common in higher stakes domains because practitioners are held more responsible for decisions (Müller et al., 2022).

A Kruskal-Wallis test found that there was a significant difference in trust in the human-machine team by years of experience ($\chi^2(3) = 15.68, p = 0.001, \eta^2 = 0.22$). Trust in the human-machine team decreased as

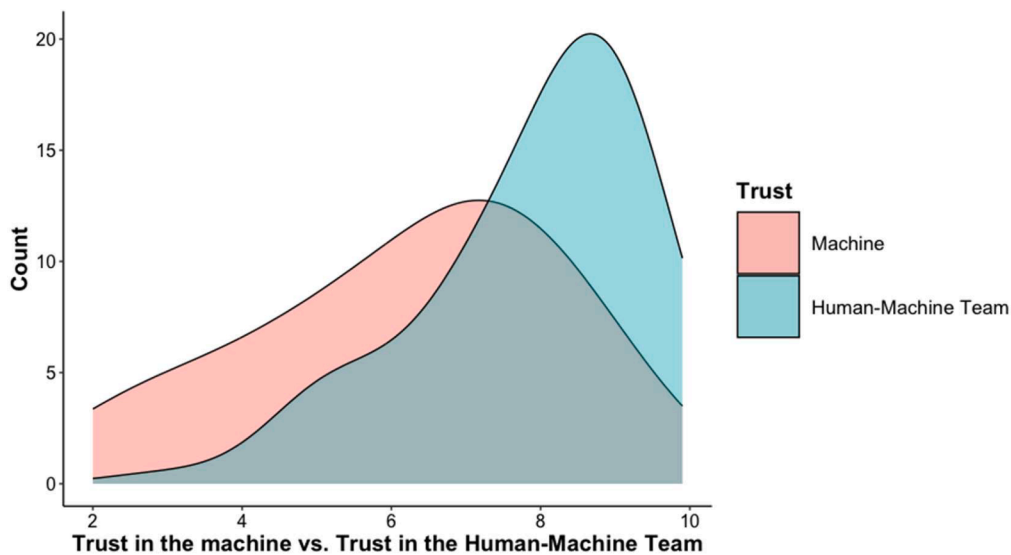


Fig. 7. A ridgeline comparison of trust ratings for the machine classification and the human-machine team. People trusted the human-machine team more than the machine's predictions alone. (Step 2b).

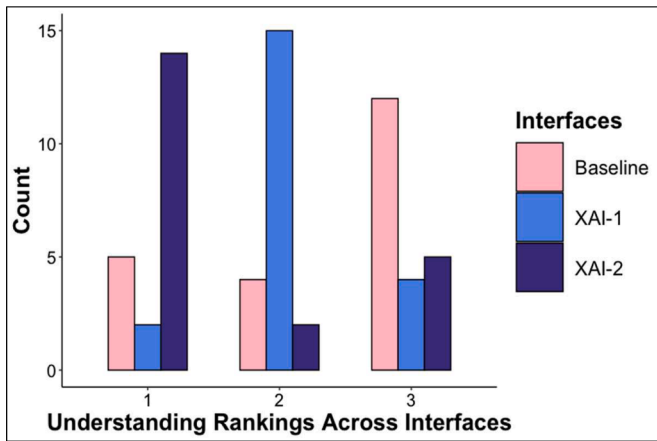


Fig. 8. A grouped bar plot comparing rankings across the interfaces. A rank of 1 indicates that an interface was considered the most easily understood. (Step 3).

the years of experience increased. The 6 month – 2 years group was the most likely to trust the team, followed by the 2 – 10 years groups, and the 10+ years group which had the lowest trust. This is interesting because years of experience was not significant for trust in the ML – it only becomes significant when a human is added to the team. This suggests that those with more experience may not find as much solace in adding a human to the decision-making process compared to less experienced personnel.

Next, we determined which features impacted the participants' trust in the human-machine team. For each interface, we trained a Random Forest Regression model to predict trust in the human-machine team using features including: years of experience, objective performance from Step 2ai, Step 2b data (including perceived agreement and understanding), and Step 3 data. The r^2 value for each model was: Baseline $r^2 = 0.87$; XAI-1 $r^2 = 0.90$; XAI-2 $r^2 = 0.90$. The Baseline and XAI models did not share the same feature importances; this suggests that adding an example-based explanation changes how people trust and think about an ML agent. For the Baseline interface, the most important feature was a measure of the participant's **nervousness** against robotics (i.e. *I feel that if I depend on robots of artificial intelligence too much, something bad might happen.*) However, for the XAI interfaces, the most important feature in predicting trust in the human-machine team was the extent to which the participant **understood** the ML's classification. This suggests that adding example-based explanations shifts participants from focusing on emotions to logical analysis while interacting with the ML agent, which would need to be further examined in future work. As people interact with ML more and more in their daily lives, it will become necessary to identify means of reducing a human's stress, nervousness, and anxiety. In terms of rhetoric, this can be thought of as a shift from pathos to logos.

4.2. Understanding

4.2.1. Understanding rankings

In Step 3, participants were asked to: Please rank each interface you understood in order of most to least. We conducted a Kruskal Wallis test and found significant differences in ranking between the interfaces ($\chi^2(2) = 9.42$, $p = 0.009$, $\eta^2 = 0.15$). Participants ranked the XAI-2 interface as the most easily understood, followed by XAI-1, and lastly the Baseline. **Given the same underlying ML agent, people felt they understood the agent the most when it provided examples from both classes.** These findings diverge from Cai et al.'s claim that having contrastive explanations is problematic (Cai et al., 2019). Our rankings indicate that having examples from both classes was more beneficial than only having examples from one class. See Fig. 8.

4.2.2. Understanding the machine's classifications

In Step 2b, participants were asked the following question: To what extent do you feel you understand why the ML made the classification it did? Overall, participants reported a moderate level of understanding. After splitting the data by interface, we found that assumptions for normality and homogeneity were met, so we conducted an ANOVA. An ANOVA reported weak differences between interfaces ($F(2, 69) = 2.520$, $p = 0.088$, $\eta p(2) = 0.068$).

A Kruskal-Wallis test found there were significant differences in understanding across years of experience ($\chi^2(3) = 18.908$, $p < 0.001$, $\eta^2 = 0.27$). The 2 – 5 years group and the 5 – 10 years group had the highest ratings, followed by the 6 month – 2 years group and 10+ years group. Participants with middling experience reported more understanding than the most and least experienced analysts.

We determined which features impacted the extent participants understood the ML agent's classifications. For each interface, we trained a Random Forest Regression model to predict understanding in the ML's classifications using features including: years of experience, objective performance from Step 2ai, Step 2b data (including perceived agreement and understanding), and Step 3 data. The r^2 value for each model was: Baseline $r^2 = 0.88$; XAI-1 $r^2 = 0.92$; XAI-2 $r^2 = 0.90$. The Baseline and XAI models did not share the same feature importances. For the Baseline interface, the most important features for predicting understanding were trust in the classification and perceived agreement. For the XAI interfaces, the **helpfulness** of showing similar examples and **trust in the human-machine team** were the **most important features in predicting understanding**.

4.2.3. Correlations between trust and understanding

We examined the correlation between the rankings for trust and understanding (Step 3). As our data was ordinal, we conducted bivariate correlation tests using Spearman's Rho. We found that there was a perfect correlation between trust and understanding for the XAI-2 interface ($r(19) = 1.00$, $p < 0.001$), and strong positive correlations for the Baseline ($r(19) = 0.904$, $p < 0.001$) and XAI-1 ($r(19) = 0.818$, $p < 0.001$) interfaces.

We also conducted Spearman's Rho tests and found positive correlations between trust and understanding when participants were evaluating individual interfaces in Step 2b. We found that there were moderately positive correlations between trust in the machine and understanding ($r(70) = 0.440$, $p < 0.001$) and trust in the human-machine team and understanding ($r(70) = 0.539$, $p < 0.001$). These findings suggest that providing explanations from both classes positively impacts both trust and understanding, and that the two are strongly related. This also can imply that **highly-technical users trust the interface more when they understand it**, and vice versa.

4.3. Perceived agreement

In Step 2b, we asked participants the following question: *How much do you agree with the ML agent's classification?* It is important to note this is perceived agreement rather than objective agreement (Step 2ai). Overall, participants reported a moderately high level of agreement with the machine ($M = 6.72$). After splitting the data by interface, we found that assumptions for normality and homogeneity were met, so we conducted an ANOVA. There were no significant differences in perceived agreement across the interfaces ($F(2, 69) = 0.832$, $p = 0.440$, $\eta p(2) = 0.075$). A Kruskal-Wallis test found that the years of experience were also not significant for perceived agreement across interfaces ($\chi^2(3) = 4.59$, $p = 0.204$, $\eta^2 = 0.065$).

We trained a Random Forest Regression model to predict perceived agreement using years of experience, objective performance from Step 2ai, Step 2b data (including trust and understanding), and Step 3 data. The r^2 value for each model was: Baseline $r^2 = 0.96$; XAI-1 $r^2 = 0.97$; XAI-2 $r^2 = 0.92$. The **most important feature for predicting perceived agreement** for all interfaces was the participant's **trust in the**

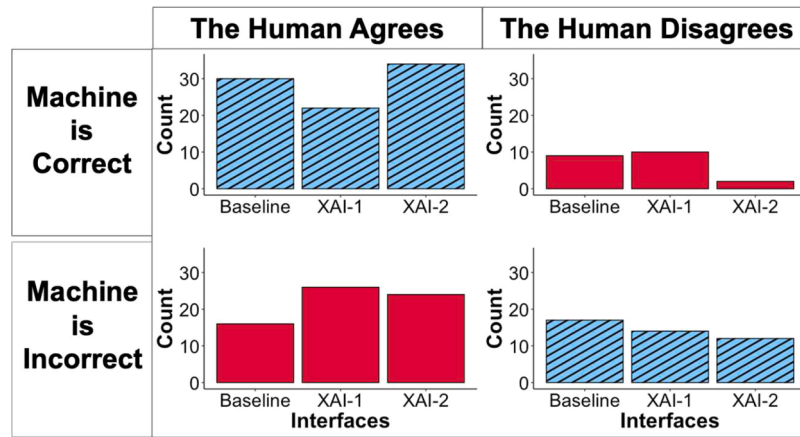


Fig. 9. Human-machine team performance across interfaces. The blue lined bars represent a human-machine team that resulted in the correct decision, while the red is a human-machine team that chose the incorrect decision. Objective Performance Category 1 is on the upper left; Category 2 on the lower left; Category 3 on the upper right; and Category 4 on the lower right.

classification.

4.4. Helpfulness

In Step 2b, for both XAI interfaces we asked participants the following question: *To what extent did showing similar cases help you determine the input's classification, regardless of whether you agreed with the ML classification?* Overall, participants had high perceptions of the machine's helpfulness. We conducted a Kruskal-Wallis test and found that the two XAI interfaces were not significantly different in helpfulness ($\chi^2(1) = 0.207, p = 0.65, \eta^2 = 0.0044$).

We also found that there were significant differences in helpfulness based on the participants' years of experience ($\chi^2(3) = 10.67, p = 0.014, \eta^2 = 0.23$). Specifically, the 2–5 years group and the 5–10 years groups had significantly higher ratings than the others, suggesting that the interfaces were more helpful for personnel with moderate levels of experience.

We trained a Random Forest Regression model to predict the helpfulness of showing similar examples in the interface using features including: years of experience, objective performance from Step 2ai, Step 2b data (including trust and understanding), and Step 3 data. The r^2 value for each model was: XAI-1 $r^2 = 0.87$; XAI-2 $r^2 = 0.83$. The **most important feature for predicting helpfulness** for both XAI models was the extent that participants **understood** why the ML agent made each classification.

4.5. Objective performance

Performance was divided into four categories:

- The human agreed with the machine and the machine was correct.
- The human agreed with the machine, but the machine was incorrect.
- The human disagreed with the machine, but the machine was correct.
- The human disagreed with the machine, and the machine was incorrect.

Any response that falls within the first or fourth categories (the human agreed with the machine and the machine was correct; the human disagreed with the machine, and the machine was incorrect) would indicate a successful human-machine team (see blue lined bars in Fig. 9). Although the fourth category has the human disagreeing with the machine, the goal is to make a correct decision, regardless of agreement. Thus, if the human overrides a machine's prediction and makes the correct decision, the team is successful. Note: This is objective

agreement recorded from Step 2ai, not perceived agreement from Step 2b.

Participants agreed with the machine 70.4 % of the time (categories 1 + 2), regardless of whether the machine's prediction was correct, with an overall success of 59.7 %. Fig. 9 summarizes the performance distributions below.

The four categories of objective performance were not significantly different across the interfaces ($\chi^2(3, N = 216) = 11.46, p = 0.075, V = 0.163$). This means that **adding in an example-based XAI interface did not reduce objective performance**. This is important because a prevalent belief in the ML community is that any concessions to human interaction will decrease performance, and therefore should not be attempted. The ML community assumes that an ideal system is a fully-autonomous system with all humans removed; however, our results indicate that providing example-based XAI can improve the human experience without reducing performance. Thus, this finding is especially important because it differs from the views of the ML community and highlights the value of improving the user experience.

The four categories of objective performance were significantly different depending on whether the case was of an anomalous or healthy portion of pipe ($\chi^2(3, N = 216) = 15.83, p = 0.001, V = 0.27$). Regardless of case type, the human and ML were both correct (Category 1) approximately 40 % of the time. It was approximately 10 % more likely for analysts to agree with the machine when it was wrong for non-anomalous cases compared to anomalous cases (non-anomalous ~35 %; anomalous ~26 %). This may point to a risk estimate by analysts who would rather agree with the ML when it says there is an anomaly, even if the case is truly non-anomalous. It was also approximately 10 % more likely for analysts to disagree with the ML when it was correct for non-anomalous cases (non-anomalous ~14 %; anomalous ~4 %). This may also point to a risk preference by analysts who would rather classify a pipeline section as anomalous (unhealthy) rather than risk an environmental disaster. It was approximately 18 % more likely for analysts to disagree with the ML when it was wrong for anomalous cases (non-anomalous ~12 %; anomalous ~30 %). This again points to the idea that analysts would rather err on the side of an anomalous classification. **In high-risk domains in which the consequences of misclassification differ by class, highly-technical analysts are likely to favor the high-risk classification, regardless of if that means agreeing or disagreeing with the ML.** This suggests that future designs should consider treating cases differently, especially if there are scenarios that involve higher risk than others.

We expected people to treat classifications differently because the risk associated with each class is different. This expands beyond the oil pipeline domain into many safety-of-life applications such as medicine,

e.g. the risks of diagnosing a patient with a disease or not. Misclassifying a non-anomalous pipeline section can lead to a large monetary loss, while a misclassification of an anomalous pipeline section can lead to an environmental disaster. In the case of other domains such as medicine, misclassification can result in endangering lives and have legal ramifications, so medical professionals have to be able to understand AI predictions and be able to take responsibility when needed (Müller et al., 2022; Caruana et al., 2015). Anomalies also occur infrequently, so personnel are more likely to be more cautious when classifying cases as anomalies.

When looking closer at whether a specific case presented to a participant was an anomaly, a difference did show up. The four categories of objective performance were significantly different across the interfaces for non-anomalous cases ($\chi^2(6, N = 119) = 14.19, p = 0.028, V = 0.244$), but not for anomalous cases ($\chi^2(6, N = 97) = 2.74, p = 0.841, V = 0.119$). This again suggests that analysts did not treat the two classifications equally.

The four categories of objective performance were not significantly different across the analysts' years of experience ($\chi^2(9, N = 216) = 6.80, p = 0.658, V = 0.102$). This suggests that performance was roughly equal regardless of how much experience an analyst had. The main difference across groups was their user experience.

We trained a Random Forest Classification model to predict objective performance using years of experience, objective agreement from Step 2ai, Step 2b data, and Step 3 data. The accuracy was 0.85. The most important feature for predicting objective performance was **objective agreement**.

4.6. Objective agreement

Accuracy is defined as whether the human-machine team ended up with the correct answer, even if the human had to override the ML's classification; *agreement* is whether the human agreed with vs. overrode the ML's classification.

Objective agreement was not significantly different for each interface ($\chi^2(6, N = 216) = 5.51, p = 0.064, V = 0.16$). People were more likely to agree with the ML, regardless of whether it was correct (70.4 % agreement; 29.6 % disagreement) (Fig. 9). Adding XAI to the interface did not change or harm whether people agreed with the ML. Objective agreement was significantly different for whether the ML was correct or not ($\chi^2(1, N = 216) = 10.18, p = 0.001, V = 0.22$). People agreed with the ML more often when the ML was correct (~57 %), and disagreed more when the ML was incorrect (~67 %). Objective agreement was not significantly different across the analysts' years of experience ($\chi^2(3, N = 216) = 3.42, p = 0.33, V = 0.13$). Less experienced analysts did not overrely on the ML to compensate for a lack of experience compared to experienced analysts.

We trained a Random Forest Classification model to predict objective agreement using years of experience, objective performance from Step 2ai, Step 2b data, and Step 3 data. The accuracy was 0.995. The most important feature for predicting objective agreement was case type (i.e. anomalous vs non-anomalous).

Interestingly, while a Spearman's Rho correlation found that there is a significant association between perceived and objective agreement ($r(214) = 0.301, p < 0.001$), it is a weak, positive correlation. This indicates that people do not accurately estimate how often they agree or disagree with the ML.

4.7. Free response factors that impact trust and understanding

During Step 2b, participants were given the option to write free-response answers regarding their interaction with the agent. For understanding, participants were prompted, "Please provide comments on what impacted your understanding of this interface." For trust, participants were prompted, "Please provide comments on what impacted your trust in this interface."

Two researchers analyzed the responses by identifying key themes from the written answers. Researchers tallied which and how many responses fit within each theme, and verified agreement. Free response questions were optional; thus, not all questions received the same number of responses. We identified the following themes: 1) priming; 2) training and education; 3) one vs. two classes in explanation; 4) interface presentation; and 5) machine errors.

4.7.1. Priming

When studying highly-technical populations at work, we have found priming to be especially important. Designers should verify that participants know the study's goal is not to replace or change their workflow, tools, or job. Two participants were concerned the ML used in the study would be integrated into their workflow without being sufficiently tested. Previous research has found that people resist new technology because they fear it will result in change when they already worked hard to adjust (Juma, 2016). This research aligns with our findings, indicating that people resist technology across different domains.

One participant claimed that trust was irrelevant because they would never use ML in their job no matter what. While this was only one participant, it is important to note that there are people with anti-ML sentiments, even among the most technical populations. This could stem from a fear of losing their job, a loss of control due to the ML making decisions, or even responsibility-authority double binds in which people could be held responsible for machine errors.

4.7.2. Training and education

Eight responses mentioned that increasing the number of cases they saw while being trained on the tool would boost trust (5 participants) or understanding (3 participants). With more examples, trust may decay at a slower rate.

Three participants had unrealistic expectations of ML, where they would "fully" trust the machine if it were 100 % accurate, but not until then. Having clear explanations about how the machine will never be perfect could help mitigate these expectations and shift trust to a more dynamic spectrum rather than a binary decision between full and no trust.

Future systems integration processes should ensure that users are familiarized with new systems through training to ensure that all users understand the system's capabilities. Training should especially involve explaining machine learning limitations such as how it is inevitable that it will make errors to mitigate unrealistic expectations.

4.7.3. One vs. two classes in explanation

Five participants mentioned that having the XAI improved either trust (1) or understanding (4). Conversely, a participant reported that having no explanations decreased their understanding.

"Having the examples certainly was the best thing, obviously the more you can rely on patterns and previous experience, the better you can make a call. It's the same in regular analysis work, if you have historical data that can give you additional info it can only benefit you in the long run, even if it may be conflicting at first."

"... it [s] very important having some examples for final classification, that could increase my trust in AI."

"Seeing what it is using as a point of reference helps in understanding why its classified as such."

"[T]here really wasn't anything to go on [with the Baseline interface], I was just looking at signals one at a time with no real understanding [of] why it did what it did. All I knew is that it didn't get them right so I had no reason to believe there weren't other errors."

Example-based explanations increased trust and understanding, while the lack of XAI decreased understanding. This is important for the adoption of ML.

4.7.4. Interface presentation

Many participants noted that the appearance and terminology of the study interfaces differed from their job’s current user interface, which impacted trust (four participants) and understanding (five participants). Training on new systems as well as being aware of resistance to change will aid designers with this issue. Future designs should ensure that the practitioners are involved in the design process to ensure that the AI interface designs match the flow of work and support them rather than hinder their work.

4.7.5. Machine errors

Four participants reported that their trust decreased after seeing the machine make an error.

“[I]t did miss a pretty obvious signal and that will make me question the validity of all the other results.”

Modern ML is based on induction; it is not possible to guarantee 100 % accuracy with any ML agent. Training users on ML basics will help this issue. Users may need to be warned that machine learning agents ‘think’ differently than humans. Specifically, similar to (Dzindolet et al., 2003), users should be informed that problems that appear obvious to humans may be difficult for ML agents to detect, and vice versa.

One participant reported that they felt skeptical after the ML made an error on a case similar to recent cases.

“[I]t got a signal wrong that looked very much like the other two signals that it got right. Because the characteristics were so similar between the ones [I] agreed with and the one [I] disagreed with, it immediately makes me think there could be other signals that [I]

think should be easily classified as anomalies that it’s getting wrong.”

The machine is able to differentiate smaller feature differences than people. As a result, the user could see the machine as a less predictable, less trusted agent. Basic ML training could mitigate trust decay.

4.8. Limitations

This work studied 24 participants recruited from one company, so the sample size was small and limited. In addition, the data was imbalanced, as there were very different numbers of participants in each experience group. Future work will verify the impact of user expertise on human-AI interaction. The participants were also from a highly technical domain, so it is unclear whether these results would generalize to a broader population. The research does not determine the effectiveness of the example-based explanations compared to other forms of explanations because it only focuses on example-based explanations. Furthermore, to mirror their real-life workflow, participants did not receive immediate feedback as to whether each decision was correct, so it is unclear whether performance and agreement rates would have differed if immediate feedback was available. Future research will expand on these findings by recruiting highly-technical populations in varied high-risk domains and examine the impact of priming and education. The impact of many factors, such as a longer time of interacting with the XAI and difficulty of cases, will also be studied in the future.

4.9. Future research

We found that using an example-based XAI shifted the main feature

Table 4
Summary of results.

Procedure Step	Topic	Summary of Results
2ai	These data were collected while the participant interacted with the agent. Healthy vs Anomalous Cases	- Objective performance was significantly different based on case type. Participants were more likely to agree with the machine when the machine predicted that a pipe was anomalous. - After splitting the data by case type, objective performance was significantly different across interfaces for ‘normal’ cases but not ‘anomalous’ cases, which suggests that participants did not treat cases the same. XAI may need to be tailored by case depending on perceived risk instead of treating all cases the same. - Participants agreed with the ML more often when the ML was correct and disagreed more when it was incorrect.
	Agreement	
2b	These data were collected immediately after the participant interacted with an interface. The participants were only asked about the particular interface they just interacted with. Trust, Understanding, Helpfulness, and Perceived Agreement in the ML’s classification	- Immediately after interacting with an interface, participants gauged trust, understanding, perceived agreement, and helpfulness in the ML approximately the same, regardless of which interface was used or the participant’s expertise. Future work will study if this is a knee-jerk reaction linked to AI anxiety. - Participants with moderate levels of experience reported more understanding in and helpfulness of the XAI than the most and least experienced groups. - Participants trusted the human-machine team (which included the participant) more than the ML alone, regardless of interface. The ability for a human or override or contribute to the decision is a more important factor for trust than the examples shown in an explanation. - More experienced participants trusted the human-machine team less than inexperienced participants. - There was a weakly positive correlation between perceived and objective agreement, suggesting that people do not accurately estimate how often they agree with the ML.
	Trust in the Human-Machine Team	
	Perceived vs Objective Agreement	
3	These data were collected at the end of the study, after the participant interacted with all interfaces. They were asked to compare the interfaces. After experiencing all interfaces and being asked to directly compare, the participants’ opinions were strong. Trust and Understanding across interfaces	- Participants trusted and understood the interface that showed examples from two classes (XAI-2) the most, followed by the interface that showed examples from one class (XAI-1), followed by no example-based explanation (Baseline). - Highly-technical users are more likely to trust the interface more when they understand it and vice versa. 1) priming; 2) training and education; 3) one vs. two classes in explanation; 4) interface presentation; and 5) machine errors. - The Baseline and XAI models did not share the same feature importances for trust in the human-machine team; this suggests that adding an example-based explanation changes how people trust and think about an ML agent. For the Baseline interface, the most important feature was a measure of AI Anxiety; for the XAI interfaces, the most important feature was the extent to which the participant understood the ML’s classification. - Many assume there is a tradeoff between performance and the human experience (Došilović et al., 2018). We show this is not true. The objective agreement and performance did not suffer from improving the human experience by adding XAI.
	Correlation between Trust and Understanding Factors that influence trust and understanding	
	Feature Importance for Trust	
	Performance vs Human Experience	

predicting participants' trust from their general nervousness of ML (negative emotions) to focusing on their understanding of the interface (logical analysis). Future work will explore more ways in which the human-agent interaction design impacts the human's emotional state. We also want to study the impact of XAI on the need for control by comparing different levels of explanation. Future work could analyze other impacts of decision-making in high-risk applications, e.g. task completion time for each case.

This research has broader societal implications because the use of artificial intelligence is growing across highly-technical domains such as the automotive industry, medical fields, law, military, and other areas such as aerospace. Ashoori et al. used hypothesized scenarios to gauge trust in AI, they found that general users preferred when there was a human involved in the training or in the decision-making (Ashoori and Weisz, 2019). More research needs to explore the impacts of human-agent design on adoption and use of ML systems by highly-technical populations and across levels of expertise within those populations.

5. Conclusion

Our main results and conclusions are shown in Table 4. Adding an example-based explanation increased trust, understanding, and helpfulness, without reducing objective performance. For our highly-technical population, we found a strong positive correlation between trust and understanding. Showing the most similar examples from two classes increased trust and understanding more than showing examples from only one class. The most important feature for predicting how helpful people found the XAI's examples was understanding. The most important feature for predicting trust in all interfaces was perceived agreement.

Our highly-technical population did not treat each classification in the domain equally; future human-ML interfaces may need to treat classifications differently. Highly-technical users may not want to use the interface without XAI regardless of accuracy because they are less likely to trust it. Thus, increasing trust is crucial for workers in technical fields to improve the adoption and use of AI.

Our findings have several practical implications. First, our findings highlight that user experience and performance are not a tradeoff; rather, user experience can be improved without harming human-machine team performance. Results from our free response section could be used to guide how new AI is introduced among technical populations. Specifically, the results could be used as a set of future design guidelines. Education and priming could help mitigate initial apprehension or unrealistic expectations for AI. While our results only focus on a small population of data analysts, our findings warrant further research on XAI's impact on user experience and performance in other high-risk domains.

CRedit authorship contribution statement

Maya Perlmutter: Formal analysis, Investigation, Validation, Visualization, Writing – original draft, Writing – review & editing, Methodology. **Ryan Gifford:** Data curation, Methodology, Software, Investigation, Writing – review & editing. **Samantha Krening:** Conceptualization, Funding acquisition, Investigation, Project administration, Supervision, Writing – original draft, Writing – review & editing, Methodology.

Declaration of competing interest

The authors have no conflicts of interests to report.

Data availability

The data that has been used is confidential.

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